Algebraic Roots & Indices

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Algebraic roots & indices

What are the laws of indices?

- **Index laws** are **rules** you can use when doing **operations** with **powers**
  - They work with both **numbers** and **algebra**
- Consider the term a^n
  - a is the **base number**
  - n is the **index**

Law	Description	How it works
$a^1 = a$	Anything to the power of 1 is itself	$x^1 = x$
$a^0 = 1$	Anything to the power of 0 is 1	$b^0 = 1$
$a^m \times a^n = a^{m+n}$	To multiply indices with the same base, add their powers	$c^3 \times c^2$ $= (c \times c \times c) \times (c \times c)$ $= c^5$
$a^m \div a^n = \frac{a^m}{a^n} = a^{m-n}$	To divide indices with the same base, subtract their powers	$d^5 \div d^2$ $= \frac{d \times d \times d \times \cancel{d} \times \cancel{d}}{\cancel{d} \times \cancel{d}}$ $= d^3$
$(a^m)^n = a^{mn}$	To raise indices to a new power, multiply their powers	$(e^3)^2$ $= (e \times e \times e) \times (e \times e \times$ $= e^6$

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	power, apply the power to both numbers, and multiply	$=j^3g^3$
$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$	To raise a fraction to a power, apply the power to both the numerator and denominator	$\left(\frac{h}{i}\right)^2 = \frac{h^2}{i^2}$
$a^{-1} = \frac{1}{a}$ $a^{-n} = \frac{1}{a^n}$	A negative power is the reciprocal	$j^{-1} = \frac{1}{j}$ $k^{-3} = \frac{1}{k^3}$

- These can be used to **simplify** expressions
  - Work out the **number** and **algebra** parts **separately**
    - $(3x^7) \times (6x^4) = (3 \times 6) \times (x^7 \times x^4) = 18x^{7+4} = 18x^{11}$
    - $\frac{6x^7}{3x^4} = \frac{6}{3} \times \frac{x^7}{x^4} = 2x^{7-4} = 2x^3$
    - $(3x^7)^2 = (3)^2 \times (x^7)^2 = 9x^{14}$

How can I solve equations with an unknown in the index?

- Write **both sides** of the equation over the **same base** number
  - Then work out what *x* should be

$$5^x = 125$$
$$5^x = 5^3$$
$$x = 3$$

- You might have to use **negative indices**

$$2^x = \frac{1}{8}$$
$$2^x = \frac{1}{2^3}$$
$$2^x = 2^{-3}$$
$$x = -3$$

Worked Example

(a) Simplify $(u^5)^5$

Answer:

Use $(a^m)^n = a^{mn}$

$$(u^5)^5 = u^{5 \times 5}$$

q^{10}

Answer:

Use $a^m \times a^n = a^{m+n}$ to simplify the numerator

$$q^2 \times q^5 = q^{2+5} = q^7$$

Use $\frac{a^m}{a^n} = a^{m-n}$ to simplify the fraction

$$\frac{q^7}{q^{10}} = q^{7-10} = q^{-3}$$

Write out both sides of the equation

$$q^x = q^{-3}$$

Both sides are now over the same base of q

So x must equal the power on the right-hand side

$$x = -3$$

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